

EPI 204 Spring 2026: Final Project Code Appendix

Andra Boca, Parker Tope, Yvonne Chen, Billy Huang, Camryn Dixon

2026-05-15

```
## Select variables of interest
nhanes_analytic <- nhanes |>
  select(
    # admin vars
    c(SEQNO, AGEYRS, AGEDIE,
      LAST_YR, LAST_MO, LAST_DAY,
      BORN_YR, BORN_MO,
      DEATH, DIE_YR, DIE_MO,
    # exposure
    NIACIN,
    # chronic conditions/comorbidities
    ANGINA, ARTHRITI, ASPIRIN , EVPREG, HTN_REP, STROKE,
    # sociodemographic variables
    MARRY, RACE, SEX, URBAN, SCHOOL,
    # lifestyle variables
    BOOZE, RECEX, AVGSMK, HEIGHT, WT)
  )

### Review covariate missingness
library(naniar)
# gg_miss_var(nhanes_analytic)

nhanes_analytic_missingness <- nhanes_analytic |>
  select(
    NIACIN,
    ANGINA,
    STROKE,
    HTN_REP,
    LAST_YR,
    BORN_YR,
    DEATH,
    URBAN,
    SCHOOL,
    ARTHRITI
  )

# Exporting table of missingness proportions for vars of interest
miss_var_summary(nhanes_analytic)

## # A tibble: 28 x 3
##   variable n_miss pct_miss
##   <chr>     <int>   <num>
## 1 AGEDIE     9113    80.9
```

```
## 2 DIE_YR      9113      80.9
## 3 DIE_MO      9113      80.9
## 4 EVPREG      6390      56.8
## 5 LAST_DAY    4153      36.9
## 6 ANGINA      2023      18.0
## 7 STROKE      2023      18.0
## 8 HTN_REP     2018      17.9
## 9 LAST_YR     2008      17.8
## 10 LAST_MO    2008      17.8
## # i 18 more rows
```

```
nhanes_analytic_missingness_table <- miss_var_summary(nhanes_analytic_missingness) |>
  as.data.frame() |>
  mutate(
    n_miss = as.numeric(n_miss),
    pct_miss = as.numeric(pct_miss)
  )
```

Produce final analytical sample keeping only participants with covariates of interest that have any missing values

```
nhanes_final <- nhanes_analytic |>
  filter(!is.na(NIACIN),
         !is.na(ANGINA),
         !is.na(STROKE),
         !is.na(HTN_REP),
         !is.na(LAST_YR),
         !is.na(BORN_YR),
         !is.na(DEATH),
         !is.na(URBAN),
         !is.na(SCHOOL),
         !is.na(ARTHRITI))
```

```
nrow(nhanes_final) #8,490 patients
```

```
## [1] 8490
```

Variable Operationalization

BMI, person-time

```
nhanes_final_2 <- nhanes_final |>
  mutate(
    # Using age as the time scale; person-time accumulates in age years
    # Time zero is age at entry
    age_tstart = AGEYRS,
    # End of follow up is censoring or death
    age_tend = if_else(!is.na(AGEDIE), AGEDIE, LAST_YR - BORN_YR),
    person_time = age_tend - age_tstart,
    event = if_else(!is.na(AGEDIE) & DEATH == 1, 1, 0),
    BMI = WT/(HEIGHT/100)^2,
    SEX = if_else(SEX == 2, 1, 0)
  )
```

```
nhanes_model <- nhanes_final_2 |>
  filter(
    !is.na(person_time),
    person_time > 0
  )
```

```
summary(nhanes_model$person_time) # min 1 and max 17 years of FU
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.00   13.00   14.00   13.37   16.00   17.00
```

Covariate List

```
# Dynamically updating lists of covariates (to automatically update all models downstream)
cov_basic <- c(
  "SEX",
  "RACE",
  "SCHOOL"
)

cov_full <- c(
  "SEX",
  "RACE",
  "SCHOOL",
  "MARRY",
  "URBAN",
  "BOOZE",
  "AVGSMK",
  "BMI",
  "ANGINA",
  "ARTHRITIS",
  "ASPIRIN",
  "HTN_REP",
  "STROKE"
)
```

Primary Analysis: Poisson Model

Generating person-year dataset

```
nhanes_final_poisson <- nhanes_model |>
  uncount(person_time, .id = "pt_year", .remove = FALSE) |>
  mutate(
    # generating individuals' age in year of follow up
    age_in_year = AGEYRS + pt_year - 1,
    # generating indicator var for death for one row per year of FU data
    death_in_year = ifelse((event == 1 & pt_year == person_time), 1, 0),
    log_pt = log(1)
  )
```

Model 1.1 Age-adjusted model

```
fit.pois1 <- glm(death_in_year ~ # exposure
  niacin_q +
  # modelling age in interval
  ns(age_in_year, df = 4),
  data=nhanes_final_poisson,
  offset=log_pt,
```

```

family=poisson(link='log'))
summary(fit.pois1)

##
## Call:
## glm(formula = death_in_year ~ niacin_q + ns(age_in_year, df = 4),
##      family = poisson(link = "log"), data = nhanes_final_poisson,
##      offset = log_pt)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -6.55022    0.39620  -16.532  < 2e-16 ***
## niacin_q2         0.04053    0.06169   0.657  0.51122
## niacin_q3         0.10666    0.06227   1.713  0.08670 .
## niacin_q4         0.20126    0.06384   3.153  0.00162 **
## ns(age_in_year, df = 4)1  2.01053    0.37672   5.337 9.45e-08 ***
## ns(age_in_year, df = 4)2  3.10903    0.23837  13.043  < 2e-16 ***
## ns(age_in_year, df = 4)3  4.01032    0.86774   4.622 3.81e-06 ***
## ns(age_in_year, df = 4)4  3.79596    0.19729  19.240  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 16201  on 113103  degrees of freedom
## Residual deviance: 14611  on 113096  degrees of freedom
## AIC: 18647
##
## Number of Fisher Scoring iterations: 8
## Poisson Model 1.1 Robust SEs
## Get robust variance-covariance matrix
robust_vcov1 <- sandwich::vcovHC(fit.pois1, type = "HC3")
tidy_model1 <- broom::tidy(fit.pois1, conf.int = TRUE, conf.level = 0.95)

# Replace standard errors with robust ones
robust_results1 <- coeftest(fit.pois1, vcov = robust_vcov1)
tidy_model1$std.error <- robust_results1[, "Std. Error"]
# Calculate RR (95% CI) with robust standard errors
tidy_poisson1 <- tidy_model1 %>%
  mutate(
    RR = exp(estimate),
    RR_low = exp(estimate - 1.96 * std.error),
    RR_high = exp(estimate + 1.96 * std.error)
  )
tidy_poisson1

## # A tibble: 8 x 10
##   term estimate std.error statistic p.value conf.low conf.high    RR RR_low
##   <chr>    <dbl>    <dbl>    <dbl>   <dbl>   <dbl>    <dbl> <dbl>
## 1 (Int~  -6.55    0.407   -16.5  2.14e-61 -7.37    -5.82  1.43e-3 6.44e-4
## 2 niac~   0.0405   0.0607   0.657  5.11e- 1 -0.0805  0.161  1.04e+0 9.24e-1
## 3 niac~   0.107    0.0614   1.71  8.67e- 2 -0.0155  0.229  1.11e+0 9.86e-1
## 4 niac~   0.201    0.0628   3.15  1.62e- 3  0.0759  0.326  1.22e+0 1.08e+0

```

```
## 5 ns(a~ 2.01      0.387      5.34 9.45e- 8 1.31      2.79 7.47e+0 3.50e+0
## 6 ns(a~ 3.11      0.243     13.0 6.98e-39 2.67      3.60 2.24e+1 1.39e+1
## 7 ns(a~ 4.01      0.893      4.62 3.81e- 6 2.39      5.80 5.52e+1 9.59e+0
## 8 ns(a~ 3.80      0.195     19.2 1.70e-82 3.42      4.19 4.45e+1 3.04e+1
## # i 1 more variable: RR_high <dbl>
```

Model 1.2: Minimally adjusted model

```
form.pois2 <- as.formula(
  paste(
    "death_in_year ~ niacin_q +",
    paste(cov_basic, collapse = " + "),
    "+ ns(age_in_year, df = 4) + offset(log_pt)"
  )
)

fit.pois2 <- glm(form.pois2,
  data=nhanes_final_poisson,
  family=poisson(link='log'))
summary(fit.pois2)

##
## Call:
## glm(formula = form.pois2, family = poisson(link = "log"), data = nhanes_final_poisson)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -5.82451    0.40977  -14.214 < 2e-16 ***
## niacin_q2       -0.04679    0.06253   -0.748 0.454221
## niacin_q3       -0.05966    0.06437   -0.927 0.353999
## niacin_q4       -0.07778    0.06829   -1.139 0.254691
## SEX             -0.66792    0.04814  -13.874 < 2e-16 ***
## RACE            -0.04948    0.05969   -0.829 0.407124
## SCHOOL          -0.05736    0.01692   -3.390 0.000698 ***
## ns(age_in_year, df = 4)1  1.93231    0.37738    5.120 3.05e-07 ***
## ns(age_in_year, df = 4)2  3.00503    0.23924   12.561 < 2e-16 ***
## ns(age_in_year, df = 4)3  3.93177    0.86892    4.525 6.04e-06 ***
## ns(age_in_year, df = 4)4  3.76338    0.19841   18.968 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 16201  on 113103  degrees of freedom
## Residual deviance: 14396  on 113093  degrees of freedom
## AIC: 18438
##
## Number of Fisher Scoring iterations: 8

## Poisson Model 1.2 Robust SEs
# Get robust variance-covariance matrix
robust_vcov2 <- sandwich::vcovHC(fit.pois2, type = "HC3")
tidy_model2 <- broom::tidy(fit.pois2, conf.int = TRUE, conf.level = 0.95)
```

```

# Replace standard errors with robust ones
robust_results2 <- coeftest(fit.pois2, vcov = robust_vcov2)
tidy_model2$std.error <- robust_results2[, "Std. Error"]
# Calculate RR (95% CI) with robust standard errors
tidy_poisson2 <- tidy_model2 %>%
  mutate(
    RR = exp(estimate),
    RR_low = exp(estimate - 1.96 * std.error),
    RR_high = exp(estimate + 1.96 * std.error)
  )
tidy_poisson2

## # A tibble: 11 x 10
##   term          estimate std.error statistic  p.value conf.low conf.high    RR
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>  <dbl>
## 1 (Intercept)   -5.82      0.421    -14.2  7.50e-46 -6.67    -5.06  2.95e-3
## 2 niacin_q2    -0.0468    0.0616    -0.748  4.54e- 1 -0.169    0.0757 9.54e-1
## 3 niacin_q3    -0.0597    0.0639    -0.927  3.54e- 1 -0.186    0.0664 9.42e-1
## 4 niacin_q4    -0.0778    0.0687    -1.14   2.55e- 1 -0.212    0.0559 9.25e-1
## 5 SEX          -0.668     0.0481   -13.9   9.10e-44 -0.762   -0.574  5.13e-1
## 6 RACE         -0.0495    0.0586    -0.829  4.07e- 1 -0.169    0.0652 9.52e-1
## 7 SCHOOL       -0.0574    0.0167    -3.39   6.98e- 4 -0.0907  -0.0244 9.44e-1
## 8 ns(age_in_y~  1.93      0.388      5.12   3.05e- 7  1.23      2.72   6.91e+0
## 9 ns(age_in_y~  3.01      0.245     12.6   3.48e-36  2.56      3.50   2.02e+1
## 10 ns(age_in_y~ 3.93      0.895      4.52   6.04e- 6  2.31      5.72   5.10e+1
## 11 ns(age_in_y~ 3.76      0.196     19.0   3.13e-80  3.38      4.16   4.31e+1
## # i 2 more variables: RR_low <dbl>, RR_high <dbl>

```

Model 1.3: Fully adjusted model

```

form.pois3 <- as.formula(
  paste(
    "death_in_year ~ niacin_q +",
    paste(cov_full, collapse = " + "),
    "+ ns(age_in_year, df = 4) + offset(log_pt)"
  )
)

fit.pois3 <- glm(form.pois3,
  data=nhanes_final_poisson,
  family=poisson(link='log'))
summary(fit.pois3)

##
## Call:
## glm(formula = form.pois3, family = poisson(link = "log"), data = nhanes_final_poisson)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -5.996472   0.446364 -13.434 < 2e-16 ***
## niacin_q2     -0.023937   0.062654  -0.382  0.70242
## niacin_q3     -0.024709   0.064722  -0.382  0.70262
## niacin_q4     -0.059843   0.068824  -0.870  0.38457
## SEX          -0.664661   0.050525 -13.155 < 2e-16 ***

```

```

## RACE -0.081498 0.062349 -1.307 0.19117
## SCHOOL -0.033983 0.017279 -1.967 0.04922 *
## MARRY 0.058841 0.020643 2.850 0.00437 **
## URBAN -0.032471 0.048242 -0.673 0.50090
## BOOZE 0.004205 0.004704 0.894 0.37135
## AVGSMK 0.019188 0.001556 12.332 < 2e-16 ***
## BMI -0.016266 0.005003 -3.252 0.00115 **
## ANGINA 0.307376 0.050622 6.072 1.26e-09 ***
## ARTHRITI -0.025099 0.048415 -0.518 0.60417
## ASPIRIN 0.093627 0.049125 1.906 0.05666 .
## HTN_REP 0.340818 0.047774 7.134 9.75e-13 ***
## STROKE 0.464359 0.090521 5.130 2.90e-07 ***
## ns(age_in_year, df = 4)1 1.900004 0.377819 5.029 4.93e-07 ***
## ns(age_in_year, df = 4)2 3.083313 0.240881 12.800 < 2e-16 ***
## ns(age_in_year, df = 4)3 4.037609 0.869247 4.645 3.40e-06 ***
## ns(age_in_year, df = 4)4 3.953025 0.200523 19.714 < 2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 16201 on 113103 degrees of freedom
## Residual deviance: 14084 on 113083 degrees of freedom
## AIC: 18146
##
## Number of Fisher Scoring iterations: 8
## Poisson Model 1.3 Robust SEs
# Get robust variance-covariance matrix
robust_vcov3 <- sandwich::vcovHC(fit.pois3, type = "HC3")
tidy_model3 <- broom::tidy(fit.pois3, conf.int = TRUE, conf.level = 0.95)

# Replace standard errors with robust ones
robust_results3 <- coeftest(fit.pois3, vcov = robust_vcov3)
tidy_model3$std.error <- robust_results3[, "Std. Error"]
# Calculate RR (95% CI) with robust standard errors
tidy_poisson3 <- tidy_model3 %>%
  mutate(
    RR = exp(estimate),
    RR_low = exp(estimate - 1.96 * std.error),
    RR_high = exp(estimate + 1.96 * std.error)
  )
tidy_poisson3

## # A tibble: 21 x 10
## term estimate std.error statistic p.value conf.low conf.high RR
## <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 (Intercept) -6.00 0.455 -13.4 3.82e-41 -6.91 -5.16 0.00249
## 2 niacin_q2 -0.0239 0.0618 -0.382 7.02e- 1 -0.147 0.0988 0.976
## 3 niacin_q3 -0.0247 0.0641 -0.382 7.03e- 1 -0.152 0.102 0.976
## 4 niacin_q4 -0.0598 0.0688 -0.870 3.85e- 1 -0.195 0.0749 0.942
## 5 SEX -0.665 0.0507 -13.2 1.59e-39 -0.764 -0.566 0.514
## 6 RACE -0.0815 0.0619 -1.31 1.91e- 1 -0.206 0.0385 0.922
## 7 SCHOOL -0.0340 0.0169 -1.97 4.92e- 2 -0.0681 -0.000311 0.967
## 8 MARRY 0.0588 0.0204 2.85 4.37e- 3 0.0177 0.0987 1.06

```

```
## 9 URBAN      -0.0325    0.0474    -0.673 5.01e- 1 -0.127    0.0618    0.968
## 10 BOOZE     0.00420    0.00488     0.894 3.71e- 1 -0.00528  0.0132    1.00
## # i 11 more rows
## # i 2 more variables: RR_low <dbl>, RR_high <dbl>
```

Model 1.4: Fully adjusted model + EMM by Sex

```
form.pois4 <- as.formula(
  paste(
    "death_in_year ~ niacin_q + niacin_q*SEX +",
    paste(cov_full, collapse = " + "),
    "+ ns(age_in_year, df = 4) + offset(log_pt)"
  )
)

fit.pois4 <- glm(form.pois4,
  data=nhanes_final_poisson,
  family=poisson(link='log'))
summary(fit.pois4)
```

```
##
## Call:
## glm(formula = form.pois4, family = poisson(link = "log"), data = nhanes_final_poisson)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -5.9604508   0.4482442  -13.297 < 2e-16 ***
## niacin_q2     -0.0379870   0.0907057   -0.419  0.67537
## niacin_q3     -0.1048332   0.0887219   -1.182  0.23737
## niacin_q4     -0.1585135   0.0882561   -1.796  0.07248 .
## SEX           -0.7533799   0.0892264   -8.443 < 2e-16 ***
## RACE          -0.0831239   0.0625537   -1.329  0.18390
## SCHOOL        -0.0327521   0.0172861   -1.895  0.05813 .
## MARRY         0.0593684   0.0206238    2.879  0.00399 **
## URBAN         -0.0273556   0.0482961   -0.566  0.57111
## BOOZE         0.0049154   0.0047039    1.045  0.29603
## AVGSMD        0.0192387   0.0015590   12.341 < 2e-16 ***
## BMI          -0.0158271   0.0050158   -3.155  0.00160 **
## ANGINA        0.3058804   0.0506275    6.042 1.52e-09 ***
## ARTHRITIS     -0.0265393   0.0484134   -0.548  0.58357
## ASPIRIN       0.0929422   0.0491160    1.892  0.05845 .
## HTN_REP      0.3411632   0.0477836    7.140 9.35e-13 ***
## STROKE        0.4644778   0.0905570    5.129 2.91e-07 ***
## ns(age_in_year, df = 4)1  1.8996930   0.3778853    5.027 4.98e-07 ***
## ns(age_in_year, df = 4)2  3.0787156   0.2409670   12.777 < 2e-16 ***
## ns(age_in_year, df = 4)3  4.0338953   0.8694628    4.640 3.49e-06 ***
## ns(age_in_year, df = 4)4  3.9483284   0.2005486   19.688 < 2e-16 ***
## niacin_q2:SEX  0.0007311   0.1255242    0.006  0.99535
## niacin_q3:SEX  0.1602200   0.1287019    1.245  0.21317
## niacin_q4:SEX  0.2921689   0.1430801    2.042  0.04115 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
```



```
##
## Null deviance: 16201 on 113103 degrees of freedom
## Residual deviance: 14078 on 113080 degrees of freedom
## AIC: 18146
##
## Number of Fisher Scoring iterations: 8

## Poisson Model 1.4 Robust SEs
# Get robust variance-covariance matrix
robust_vcov4 <- sandwich::vcovHC(fit.pois4, type = "HC3")
tidy_model4 <- broom::tidy(fit.pois4, conf.int = TRUE, conf.level = 0.95)

# Replace standard errors with robust ones
robust_results4 <- coeftest(fit.pois4, vcov = robust_vcov4)
tidy_model4$std.error <- robust_results4[, "Std. Error"]
# Calculate RR (95% CI) with robust standard errors
tidy_poisson4 <- tidy_model4 %>%
  mutate(
    RR = exp(estimate),
    RR_low = exp(estimate - 1.96 * std.error),
    RR_high = exp(estimate + 1.96 * std.error)
  )
tidy_poisson4
```

```
## # A tibble: 24 x 10
##   term      estimate std.error statistic  p.value conf.low conf.high   RR
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl> <dbl>
## 1 (Intercept) -5.96      0.457    -13.3  2.40e-40 -6.88    -5.12  0.00258
## 2 niacin_q2   -0.0380   0.0887    -0.419  6.75e- 1 -0.215    0.140  0.963
## 3 niacin_q3   -0.105    0.0867    -1.18  2.37e- 1 -0.278    0.0700 0.900
## 4 niacin_q4   -0.159    0.0868    -1.80  7.25e- 2 -0.331    0.0156 0.853
## 5 SEX         -0.753    0.0883    -8.44  3.08e-17 -0.928   -0.578 0.471
## 6 RACE        -0.0831   0.0623    -1.33  1.84e- 1 -0.208    0.0373 0.920
## 7 SCHOOL      -0.0328   0.0169    -1.89  5.81e- 2 -0.0668   0.000933 0.968
## 8 MARRY        0.0594   0.0204     2.88  3.99e- 3  0.0183    0.0992 1.06
## 9 URBAN       -0.0274   0.0475    -0.566 5.71e- 1 -0.122    0.0670 0.973
## 10 BOOZE       0.00492   0.00487     1.04  2.96e- 1 -0.00457  0.0139 1.00
## # i 14 more rows
## # i 2 more variables: RR_low <dbl>, RR_high <dbl>
```

```
poisson_emm_results <- data.frame(
  coefs = robust_results4[c("niacin_q2", "niacin_q3", "niacin_q4", "SEX", "niacin_q2:SEX", "niacin_q3:SEX", "niacin_q4:SEX"),
  se = robust_results4[c("niacin_q2", "niacin_q3", "niacin_q4", "SEX", "niacin_q2:SEX", "niacin_q3:SEX", "niacin_q4:SEX"),
)
poisson_emm_results$variance <- (poisson_emm_results$se)^2
v <- robust_vcov4
```

Table 3: HRs for FEMALES (SEX == 1)

```
female.hr <- data.frame(
  niacin_q = c("niacin_q2", "niacin_q3", "niacin_q4"),
  variance = c(
    v["niacin_q2", "niacin_q2"] + v["niacin_q2:SEX", "niacin_q2:SEX"] + 2*v["niacin_q2", "niacin_q2:SEX"] +
    v["niacin_q3", "niacin_q3"] + v["niacin_q3:SEX", "niacin_q3:SEX"] + 2*v["niacin_q3", "niacin_q3:SEX"] +
    v["niacin_q4", "niacin_q4"] + v["niacin_q4:SEX", "niacin_q4:SEX"] + 2*v["niacin_q4", "niacin_q4:SEX"]
  )
)
```

```

coef = c(
  robust_results4[c("niacin_q2"), "Estimate"] + robust_results4[c("niacin_q2:SEX"), "Estimate"],
  robust_results4[c("niacin_q3"), "Estimate"] + robust_results4[c("niacin_q3:SEX"), "Estimate"],
  robust_results4[c("niacin_q4"), "Estimate"] + robust_results4[c("niacin_q4:SEX"), "Estimate"],
  sex = c("female", "female", "female")
)

female.hr <- female.hr |>
  mutate(
    hr = exp(coef),
    lcl = exp(coef - (qnorm(0.975)*sqrt(variance))),
    ucl = exp(coef + (qnorm(0.975)*sqrt(variance)))
  )

```

Table 3: HRs for MALEs (SEX == 0)

```

male.hr <- data.frame(
  niacin_q = c("niacin_q2", "niacin_q3", "niacin_q4"),
  variance = c(
    v["niacin_q2", "niacin_q2"],
    v["niacin_q3", "niacin_q3"],
    v["niacin_q4", "niacin_q4"]),
  coef = c(
    robust_results4[c("niacin_q2"), "Estimate"],
    robust_results4[c("niacin_q3"), "Estimate"],
    robust_results4[c("niacin_q4"), "Estimate"]),
  sex = c("male", "male", "male")
)

male.hr <- male.hr |>
  mutate(
    hr = exp(coef),
    lcl = exp(coef - (qnorm(0.975)*sqrt(variance))),
    ucl = exp(coef + (qnorm(0.975)*sqrt(variance)))
  )

```

Stratified Analysis

```

nhanes_final_poisson_male <- nhanes_final_poisson |>
  filter(SEX == 0)

table(nhanes_final_poisson_male$niacin_q)

```

Stratifying data

```

##
##      1      2      3      4
## 6111 9627 13953 20614

nhanes_final_poisson_female <- nhanes_final_poisson |>
  filter(SEX == 1)
table(nhanes_final_poisson_female$niacin_q)

##

```

```
##      1      2      3      4
## 21732 18459 14371 8237
```

```
cov_full_stratified <- c(
  "RACE",
  "SCHOOL",
  "MARRY",
  "URBAN",
  "BOOZE",
  "AVGSMK",
  "BMI",
  "ANGINA",
  "ARTHRITI",
  "ASPIRIN",
  "HTN_REP",
  "STROKE"
)
```

Full covariate set for stratified models

Model 1.5: Fully-adjusted model stratified by sex - male

```
form.pois5 <- as.formula(
  paste(
    "death_in_year ~ niacin_q + ",
    paste(cov_full_stratified, collapse = " + "),
    "+ ns(age_in_year, df = 4) + offset(log_pt)"
  )
)

fit.pois5 <- glm(form.pois5,
  data=nhanes_final_poisson_male,
  family=poisson(link='log'))
summary(fit.pois5)
```

```
##
## Call:
## glm(formula = form.pois5, family = poisson(link = "log"), data = nhanes_final_poisson_male)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -5.679544    0.598483  -9.490 < 2e-16 ***
## niacin_q2      -0.042589    0.091233  -0.467  0.640632
## niacin_q3      -0.113618    0.089551  -1.269  0.204527
## niacin_q4      -0.161350    0.089592  -1.801  0.071713 .
## RACE           -0.188343    0.082731  -2.277  0.022812 *
## SCHOOL         -0.038969    0.022415  -1.739  0.082116 .
## MARRY           0.095063    0.025882   3.673  0.000240 ***
## URBAN          -0.039383    0.062299  -0.632  0.527279
## BOOZE           0.004240    0.005098   0.832  0.405567
## AVGSMK          0.017066    0.001896   9.000 < 2e-16 ***
## BMI            -0.023947    0.007501  -3.192  0.001412 **
## ANGINA          0.385440    0.066479   5.798  6.71e-09 ***
```

```
## ARTHRITI          -0.016232    0.064098   -0.253 0.800090
## ASPIRIN           0.064714    0.066125    0.979 0.327744
## HTN_REP           0.304282    0.062093    4.900 9.56e-07 ***
## STROKE            0.464189    0.117942    3.936 8.29e-05 ***
## ns(age_in_year, df = 4)1 1.900836  0.498587    3.812 0.000138 ***
## ns(age_in_year, df = 4)2 3.147293  0.319937    9.837 < 2e-16 ***
## ns(age_in_year, df = 4)3 3.937418  1.144808    3.439 0.000583 ***
## ns(age_in_year, df = 4)4 3.760830  0.275733   13.639 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 8872.6 on 50304 degrees of freedom
## Residual deviance: 7726.6 on 50285 degrees of freedom
## AIC: 10133
##
## Number of Fisher Scoring iterations: 7

## Poisson Model 1.5 Robust SEs
# Get robust variance-covariance matrix
robust_vcov5 <- sandwich::vcovHC(fit.pois5, type = "HC3")
tidy_model5 <- broom::tidy(fit.pois5, conf.int = TRUE, conf.level = 0.95)

# Replace standard errors with robust ones
robust_results5 <- coeftest(fit.pois5, vcov = robust_vcov5)
tidy_model5$std.error <- robust_results5[, "Std. Error"]
# Calculate RR (95% CI) with robust standard errors
tidy_poisson5 <- tidy_model5 %>%
  mutate(
    RR = exp(estimate),
    RR_low = exp(estimate - 1.96 * std.error),
    RR_high = exp(estimate + 1.96 * std.error)
  )
tidy_poisson5

## # A tibble: 20 x 10
##   term      estimate std.error statistic  p.value conf.low conf.high    RR
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>  <dbl>
## 1 (Intercept) -5.68      0.580     -9.49 2.31e-21 -6.92     -4.56  3.42e-3
## 2 niacin_q2  -0.0426    0.0892     -0.467 6.41e- 1 -0.221     0.137  9.58e-1
## 3 niacin_q3  -0.114     0.0875     -1.27 2.05e- 1 -0.288     0.0628 8.93e-1
## 4 niacin_q4  -0.161     0.0880     -1.80 7.17e- 2 -0.336     0.0153 8.51e-1
## 5 RACE       -0.188     0.0815     -2.28 2.28e- 2 -0.354    -0.0300 8.28e-1
## 6 SCHOOL     -0.0390    0.0220     -1.74 8.21e- 2 -0.0833    0.00462 9.62e-1
## 7 MARRY       0.0951    0.0250      3.67 2.40e- 4  0.0432     0.145  1.10e+0
## 8 URBAN      -0.0394    0.0609     -0.632 5.27e- 1 -0.162     0.0824 9.61e-1
## 9 BOOZE       0.00424    0.00536     0.832 4.06e- 1 -0.00606    0.0139 1.00e+0
## 10 AVGSMK     0.0171    0.00167     9.00 2.26e-19  0.0133     0.0207 1.02e+0
## 11 BMI       -0.0239    0.00762     -3.19 1.41e- 3 -0.0387    -0.00932 9.76e-1
## 12 ANGINA     0.385     0.0661      5.80 6.71e- 9  0.254     0.515  1.47e+0
## 13 ARTHRITI  -0.0162    0.0629     -0.253 8.00e- 1 -0.143     0.109  9.84e-1
## 14 ASPIRIN    0.0647    0.0638      0.979 3.28e- 1 -0.0659     0.193  1.07e+0
## 15 HTN_REP    0.304     0.0604      4.90 9.56e- 7  0.182     0.426  1.36e+0
## 16 STROKE     0.464     0.114      3.94 8.29e- 5  0.226     0.689  1.59e+0
```

```
## 17 ns(age_in_y~ 1.90      0.484      3.81 1.38e- 4 0.994      2.96      6.69e+0
## 18 ns(age_in_y~ 3.15      0.310      9.84 7.78e-23 2.56      3.82      2.33e+1
## 19 ns(age_in_y~ 3.94      1.10       3.44 5.83e- 4 1.83      6.34      5.13e+1
## 20 ns(age_in_y~ 3.76      0.270     13.6 2.34e-42 3.23      4.31      4.30e+1
## # i 2 more variables: RR_low <dbl>, RR_high <dbl>
```

Model 1.6: Fully-adjusted model stratified by sex - female

```
form.pois6 <- as.formula(
  paste(
    "death_in_year ~ niacin_q + ",
    paste(cov_full_stratified, collapse = " + "),
    "+ ns(age_in_year, df = 4) + offset(log_pt)"
  )
)

fit.pois6 <- glm(form.pois6,
  data=nhanes_final_poisson_female,
  family=poisson(link='log'))
summary(fit.pois6)

##
## Call:
## glm(formula = form.pois6, family = poisson(link = "log"), data = nhanes_final_poisson_female)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -7.040440   0.682752 -10.312 < 2e-16 ***
## niacin_q2      -0.043684   0.087859  -0.497  0.61904
## niacin_q3       0.055336   0.095657   0.578  0.56294
## niacin_q4       0.135265   0.115507   1.171  0.24158
## RACE            0.082518   0.096596   0.854  0.39296
## SCHOOL        -0.016425   0.027585  -0.595  0.55155
## MARRY          -0.005090   0.034039  -0.150  0.88114
## URBAN          -0.011199   0.077062  -0.145  0.88445
## BOOZE           0.008845   0.012303   0.719  0.47217
## AVGSMD         0.023830   0.002764   8.623 < 2e-16 ***
## BMI            -0.009033   0.006845  -1.320  0.18696
## ANGINA          0.204177   0.078755   2.593  0.00953 **
## ARTHRITIS      -0.034722   0.074219  -0.468  0.63990
## ASPIRIN         0.119617   0.073923   1.618  0.10564
## HTN_REP        0.388888   0.075856   5.127 2.95e-07 ***
## STROKE         0.460574   0.142185   3.239  0.00120 **
## ns(age_in_year, df = 4)1  1.928732   0.587596   3.282  0.00103 **
## ns(age_in_year, df = 4)2  3.105452   0.376426   8.250 < 2e-16 ***
## ns(age_in_year, df = 4)3  4.296474   1.357135   3.166  0.00155 **
## ns(age_in_year, df = 4)4  4.146720   0.300572  13.796 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 7161.6  on 62798  degrees of freedom
## Residual deviance: 6330.0  on 62779  degrees of freedom
```

```

## AIC: 8024
##
## Number of Fisher Scoring iterations: 8
## Poisson Model 1.6 Robust SEs
# Get robust variance-covariance matrix
robust_vcov6 <- sandwich::vcovHC(fit.pois6, type = "HC3")
tidy_model6 <- broom::tidy(fit.pois6, conf.int = TRUE, conf.level = 0.95)

# Replace standard errors with robust ones
robust_results6 <- coeftest(fit.pois6, vcov = robust_vcov6)
tidy_model6$std.error <- robust_results6[, "Std. Error"]
# Calculate RR (95% CI) with robust standard errors
tidy_poisson6 <- tidy_model6 %>%
  mutate(
    RR = exp(estimate),
    RR_low = exp(estimate - 1.96 * std.error),
    RR_high = exp(estimate + 1.96 * std.error)
  )
tidy_poisson6

## # A tibble: 20 x 10
##   term      estimate std.error statistic  p.value conf.low conf.high    RR
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>  <dbl>
## 1 (Intercept) -7.04      0.744    -10.3  6.23e-25 -8.48    -5.79    8.76e-4
## 2 niacin_q2   -0.0437    0.0888    -0.497  6.19e- 1 -0.217    0.128    9.57e-1
## 3 niacin_q3    0.0553    0.0948     0.578  5.63e- 1 -0.134    0.241    1.06e+0
## 4 niacin_q4    0.135     0.114     1.17   2.42e- 1 -0.0955    0.358    1.14e+0
## 5 RACE         0.0825    0.0977     0.854  3.93e- 1 -0.112    0.267    1.09e+0
## 6 SCHOOL      -0.0164    0.0271    -0.595  5.52e- 1 -0.0709    0.0372    9.84e-1
## 7 MARRY       -0.00509    0.0341    -0.150  8.81e- 1 -0.0733    0.0602    9.95e-1
## 8 URBAN       -0.0112    0.0770    -0.145  8.84e- 1 -0.163    0.139    9.89e-1
## 9 BOOZE        0.00885    0.0117     0.719  4.72e- 1 -0.0170    0.0311    1.01e+0
## 10 AVGSMK      0.0238    0.00251     8.62  6.52e-18  0.0182    0.0291    1.02e+0
## 11 BMI        -0.00903    0.00784    -1.32  1.87e- 1 -0.0226    0.00422    9.91e-1
## 12 ANGINA      0.204     0.0778     2.59  9.53e- 3  0.0484    0.357    1.23e+0
## 13 ARTHRITI   -0.0347    0.0748    -0.468  6.40e- 1 -0.180    0.111    9.66e-1
## 14 ASPIRIN     0.120     0.0738     1.62  1.06e- 1 -0.0260    0.264    1.13e+0
## 15 HTN_REP     0.389     0.0755     5.13  2.95e- 7  0.240    0.538    1.48e+0
## 16 STROKE      0.461     0.141     3.24  1.20e- 3  0.171    0.730    1.58e+0
## 17 ns(age_in_y~ 1.93     0.648     3.28  1.03e- 3  0.874    3.19    6.88e+0
## 18 ns(age_in_y~ 3.11     0.412     8.25  1.59e-16  2.43     3.91    2.23e+1
## 19 ns(age_in_y~ 4.30     1.52      3.17  1.55e- 3  1.83     7.18    7.34e+1
## 20 ns(age_in_y~ 4.15     0.304    13.8  2.69e-43  3.57     4.76    6.32e+1
## # i 2 more variables: RR_low <dbl>, RR_high <dbl>

# LRT #1: Compare 1.3.1 [null] to 1.3.2 [alternative]
lrtest(fit.pois1.3.1, fit.pois1.3.2)

## Likelihood ratio test
##
## Model 1: death_in_year ~ niacin_q + SEX + RACE + SCHOOL + MARRY + URBAN +
##   BOOZE + AVGSMK + BMI + ANGINA + ARTHRITI + ASPIRIN + HTN_REP +
##   STROKE + offset(log_pt)
## Model 2: death_in_year ~ niacin_q + SEX + RACE + SCHOOL + MARRY + URBAN +

```

```

##      BOOZE + AVGSMK + BMI + ANGINA + ARTHRITI + ASPIRIN + HTN_REP +
##      STROKE + age_in_year + offset(log_pt)
##      #Df  LogLik Df  Chisq Pr(>Chisq)
## 1   17 -9731.7
## 2   18 -9057.7   1  1348   < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# LRT #2: Compare 1.3.1 [null] to 1.3.3 [alternative]
lrtest(fit.pois1.3.1, fit.pois1.3.3)

## Likelihood ratio test
##
## Model 1: death_in_year ~ niacin_q + SEX + RACE + SCHOOL + MARRY + URBAN +
##      BOOZE + AVGSMK + BMI + ANGINA + ARTHRITI + ASPIRIN + HTN_REP +
##      STROKE + offset(log_pt)
## Model 2: death_in_year ~ niacin_q + SEX + RACE + SCHOOL + MARRY + URBAN +
##      BOOZE + AVGSMK + BMI + ANGINA + ARTHRITI + ASPIRIN + HTN_REP +
##      STROKE + ns(age_in_year, df = 4) + offset(log_pt)
##      #Df  LogLik Df  Chisq Pr(>Chisq)
## 1   17 -9731.7
## 2   21 -9051.9   4 1359.5   < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# LRT #3: Compare 1.3.2 [null] to 1.3.3 [alternative]
lrtest(fit.pois1.3.2, fit.pois1.3.3)

## Likelihood ratio test
##
## Model 1: death_in_year ~ niacin_q + SEX + RACE + SCHOOL + MARRY + URBAN +
##      BOOZE + AVGSMK + BMI + ANGINA + ARTHRITI + ASPIRIN + HTN_REP +
##      STROKE + age_in_year + offset(log_pt)
## Model 2: death_in_year ~ niacin_q + SEX + RACE + SCHOOL + MARRY + URBAN +
##      BOOZE + AVGSMK + BMI + ANGINA + ARTHRITI + ASPIRIN + HTN_REP +
##      STROKE + ns(age_in_year, df = 4) + offset(log_pt)
##      #Df  LogLik Df  Chisq Pr(>Chisq)
## 1   18 -9057.7
## 2   21 -9051.9   3 11.463   0.009469 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Model 1.3.4
form.pois1.3.4 <- as.formula(
  paste(
    "death_in_year ~ NIACIN +",
    paste(cov_full, collapse = " + "),
    "+ ns(age_in_year, df = 4) + offset(log_pt)"
  )
)

fit.pois1.3.4 <- glm(form.pois1.3.4,
  data=nhanes_final_poisson,
  family=poisson(link='log'))

## Model 1.3.5

```

```
form.pois1.3.5 <- as.formula(
  paste(
    "death_in_year ~ ns(NIACIN, df = 4) +",
    paste(cov_full, collapse = " + "),
    "+ ns(age_in_year, df = 4) + offset(log_pt)"
  )
)

fit.pois1.3.5 <- glm(form.pois1.3.5,
  data=nhanes_final_poisson,
  family=poisson(link='log'))
```

```
# LRT #4: Compare 1.3.4 [null] to 1.3.5 [alternative]
lrtest(fit.pois1.3.4, fit.pois1.3.5)
```

```
## Likelihood ratio test
##
## Model 1: death_in_year ~ NIACIN + SEX + RACE + SCHOOL + MARRY + URBAN +
##      BOOZE + AVGSMK + BMI + ANGINA + ARTHRITI + ASPIRIN + HTN_REP +
##      STROKE + ns(age_in_year, df = 4) + offset(log_pt)
## Model 2: death_in_year ~ ns(NIACIN, df = 4) + SEX + RACE + SCHOOL + MARRY +
##      URBAN + BOOZE + AVGSMK + BMI + ANGINA + ARTHRITI + ASPIRIN +
##      HTN_REP + STROKE + ns(age_in_year, df = 4) + offset(log_pt)
##      #Df  LogLik Df  Chisq Pr(>Chisq)
## 1   19 -9052.3
## 2   22 -9049.8  3  5.0502    0.1682
```

Cox Proportional Hazards Sensitivity Models

```
# Age adjusted model
fit.cox.unadj <- coxph(
  Surv(age_tstart, age_tend, event) ~ niacin_q,
  data = nhanes_model,
  ties = 'efron'
)
summary(fit.cox.unadj)

## Call:
## coxph(formula = Surv(age_tstart, age_tend, event) ~ niacin_q,
##      data = nhanes_model, ties = "efron")
##
##      n= 8461, number of events= 2010
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## niacin_q2 0.04120   1.04206  0.06169  0.668  0.50423
## niacin_q3 0.10888   1.11503  0.06227  1.748  0.08039 .
## niacin_q4 0.20470   1.22715  0.06386  3.206  0.00135 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## niacin_q2      1.042      0.9596   0.9234   1.176
## niacin_q3      1.115      0.8968   0.9869   1.260
## niacin_q4      1.227      0.8149   1.0828   1.391
```



```
##
## Concordance= 0.516 (se = 0.007 )
## Likelihood ratio test= 11.43 on 3 df, p=0.01
## Wald test = 11.57 on 3 df, p=0.009
## Score (logrank) test = 11.6 on 3 df, p=0.009

cox.age <- tidy(
  fit.cox.unadj,
  exponentiate = TRUE,
  conf.int = TRUE
) %>%
  select(term, estimate, conf.low, conf.high, p.value) %>%
  rename(
    HR = estimate,
    CI_lower = conf.low,
    CI_upper = conf.high,
    p_value = p.value
  )

# Minimally adjusted model
fit.cox.min.adj <- coxph(
  as.formula(
    paste(
      "Surv(age_tstart, age_tend, event) ~ niacin_q +",
      paste(cov_basic, collapse = " + ")
    )
  ),
  data = nhanes_model,
  ties = 'efron'
)
summary(fit.cox.min.adj)

## Call:
## coxph(formula = as.formula(paste("Surv(age_tstart, age_tend, event) ~ niacin_q +",
## paste(cov_basic, collapse = " + "))), data = nhanes_model,
## ties = "efron")
##
## n= 8461, number of events= 2010
##
##          coef exp(coef) se(coef)      z Pr(>|z|)
## niacin_q2 -0.04789   0.95324  0.06254  -0.766 0.443867
## niacin_q3 -0.06109   0.94074  0.06439  -0.949 0.342765
## niacin_q4 -0.07999   0.92312  0.06833  -1.171 0.241744
## SEX       -0.67909   0.50708  0.04817 -14.099 < 2e-16 ***
## RACE      -0.05153   0.94978  0.05970  -0.863 0.388059
## SCHOOL    -0.05831   0.94336  0.01692  -3.446 0.000568 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##          exp(coef) exp(-coef) lower .95 upper .95
## niacin_q2    0.9532    1.049    0.8433    1.0776
## niacin_q3    0.9407    1.063    0.8292    1.0673
## niacin_q4    0.9231    1.083    0.8074    1.0554
## SEX          0.5071    1.972    0.4614    0.5573
## RACE          0.9498    1.053    0.8449    1.0677
## SCHOOL       0.9434    1.060    0.9126    0.9752
```

```
##
## Concordance= 0.602 (se = 0.007 )
## Likelihood ratio test= 233.1 on 6 df, p=<2e-16
## Wald test = 228.4 on 6 df, p=<2e-16
## Score (logrank) test = 236.3 on 6 df, p=<2e-16

cox.minimal <- tidy(
  fit.cox.min.adj,
  exponentiate = TRUE,
  conf.int = TRUE
) %>%
select(term, estimate, conf.low, conf.high, p.value) %>%
rename(
  HR = estimate,
  CI_lower = conf.low,
  CI_upper = conf.high,
  p_value = p.value
)

# Fully adjusted model
fit.cox.full.adj <- coxph(
  as.formula(
    paste(
      "Surv(age_tstart, age_tend, event) ~ niacin_q +",
      paste(cov_full, collapse = " + ")
    )
  ),
  data = nhanes_model,
  ties = 'efron'
)
summary(fit.cox.full.adj)

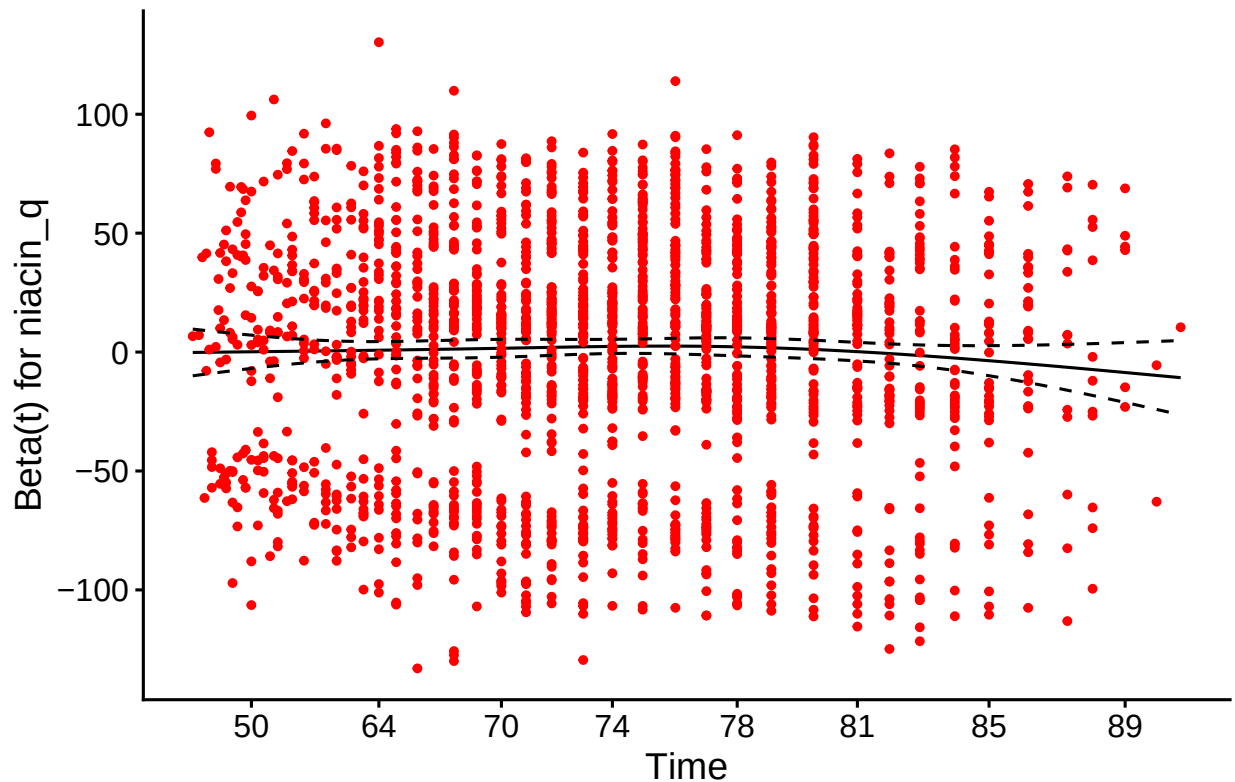
## Call:
## coxph(formula = as.formula(paste("Surv(age_tstart, age_tend, event) ~ niacin_q +",
## paste(cov_full, collapse = " + "))), data = nhanes_model,
## ties = "efron")
##
## n= 8461, number of events= 2010
##
##      coef exp(coef) se(coef)      z Pr(>|z|)
## niacin_q2 -0.025647  0.974679  0.062670  -0.409 0.682364
## niacin_q3 -0.026086  0.974252  0.064750  -0.403 0.687047
## niacin_q4 -0.062615  0.939305  0.068869  -0.909 0.363253
## SEX      -0.677271  0.508001  0.050562 -13.395 < 2e-16 ***
## RACE      -0.084290  0.919165  0.062384  -1.351 0.176648
## SCHOOL    -0.034313  0.966269  0.017277  -1.986 0.047032 *
## MARRY      0.059334  1.061129  0.020645   2.874 0.004053 **
## URBAN     -0.032598  0.967928  0.048267  -0.675 0.499450
## BOOZE      0.004349  1.004358  0.004717   0.922 0.356560
## AVGSMD     0.019540  1.019732  0.001554  12.574 < 2e-16 ***
## BMI       -0.016885  0.983257  0.005013  -3.368 0.000756 ***
## ANGINA     0.313254  1.367869  0.050630   6.187 6.13e-10 ***
## ARTHRITI  -0.024345  0.975948  0.048437  -0.503 0.615232
## ASPIRIN    0.095941  1.100694  0.049125   1.953 0.050823 .
## HTN_REP    0.348319  1.416683  0.047765   7.292 3.05e-13 ***
## STROKE     0.478503  1.613657  0.090570   5.283 1.27e-07 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      exp(coef) exp(-coef) lower .95 upper .95
## niacin_q2      0.9747      1.0260      0.8620      1.1021
## niacin_q3      0.9743      1.0264      0.8581      1.1061
## niacin_q4      0.9393      1.0646      0.8207      1.0750
## SEX            0.5080      1.9685      0.4601      0.5609
## RACE           0.9192      1.0879      0.8134      1.0387
## SCHOOL         0.9663      1.0349      0.9341      0.9995
## MARRY          1.0611      0.9424      1.0191      1.1049
## URBAN          0.9679      1.0331      0.8806      1.0640
## BOOZE          1.0044      0.9957      0.9951      1.0137
## AVGSMD         1.0197      0.9806      1.0166      1.0228
## BMI            0.9833      1.0170      0.9736      0.9930
## ANGINA         1.3679      0.7311      1.2386      1.5106
## ARTHRITIS      0.9759      1.0246      0.8876      1.0731
## ASPIRIN        1.1007      0.9085      0.9997      1.2119
## HTN_REP        1.4167      0.7059      1.2901      1.5557
## STROKE         1.6137      0.6197      1.3512      1.9271
##
## Concordance= 0.658 (se = 0.007 )
## Likelihood ratio test= 556.7 on 16 df,  p=<2e-16
## Wald test              = 618.9 on 16 df,  p=<2e-16
## Score (logrank) test = 640.5 on 16 df,  p=<2e-16

cox.full <- tidy(
  fit.cox.full.adj,
  exponentiate = TRUE,
  conf.int = TRUE
) %>%
  select(term, estimate, conf.low, conf.high, p.value) %>%
  rename(
    HR = estimate,
    CI_lower = conf.low,
    CI_upper = conf.high,
    p_value = p.value
  )

#checking Schoenfeld Residuals
check.pha<-cox.zph(fit.cox.full.adj)
ggcoxzph(check.pha[1])
```

Schoenfeld Individual Test p: 0.1294



```
table.1<-nhanes_final_2 %>%
  arrange(niacin_q) %>%
  mutate(niacin_cat=factor(niacin_q,levels=c(1,2,3,4)),
    sex=factor(SEX, labels=c("male", "female"), levels=c(1,2)),
    age= AGEYRS,
    angina=factor(ANGINA, levels=c(0,1)),
    arthritis = factor(ARTHRITI, levels=c(0,1)),
    aspirin = factor(ASPIRIN, levels=c(0,1)),
    ever_pregnant=factor(EVPREG,levels=c(0,1)), # might need to show it only among females
    hypertension=factor(HTN_REP,levels=c(0,1)),
    marital_status=factor(MARRY,
      labels=c("Under 17", "Married", "Widowed",
        "Divorced", "Separated", "Never Married",
        "Blank, but applicable"),
      levels=c(1,2,3,4,5,6,8)),
    race=factor(RACE,
      labels=c("White","Black","Other"),
      levels=c(1,2,3)),
    urban=factor(URBAN,
      labels=c("Yes", "No"), levels=c(1,2)),
    school=factor(SCHOOL,
      labels=c("Elementary", "Some high school",
        "High school", "Some college", "College"),
      levels=c(1,2,3,4,5)),
    alcohol=as.numeric(BOOZE),
    recex=factor(RECEX, labels=c("High","Medium","Low"),levels=c(1,2,3)),
```

```

smoking=as.numeric(AVGSMK),
bmi=as.numeric(BMI))

vars <- (c("sex", "race", "age", "angina", "arthritis",
          "hypertension", "aspirin",
          "school",
          "marital_status", "urban",
          "alcohol", "recex", "smoking",
          "bmi")) # include more covariates as needed

# Table 1 needs complete case

t1 <- CreateTableOne(vars = vars, data = table.1, strata = as.factor("niacin_q"))
options(digits = 2)
print(t1, showAllLevels = TRUE)

```

```

##              Stratified by niacin_q
##              level                1                2
##  n              2123              2121
##  sex (%)        male            1596 (100.0)      1339 (100.0)
##              female              0 ( 0.0)         0 ( 0.0)
##  race (%)       White            1728 ( 81.4)      1864 ( 87.9)
##              Black              359 ( 16.9)        228 ( 10.7)
##              Other              36 (  1.7)         29 (  1.4)
##  age (mean (SD)) 57.48 (12.73)  55.59 (12.99)
##  angina (%)      0            1549 ( 73.0)      1638 ( 77.2)
##              1              574 ( 27.0)         483 ( 22.8)
##  arthritis (%)   0            1300 ( 61.2)      1407 ( 66.3)
##              1              823 ( 38.8)         714 ( 33.7)
##  hypertension (%) 0            1198 ( 56.4)      1363 ( 64.3)
##              1              925 ( 43.6)         758 ( 35.7)
##  aspirin (%)     0            1398 ( 65.9)      1446 ( 68.2)
##              1              725 ( 34.1)         675 ( 31.8)
##  school (%)      Elementary      811 ( 38.2)         617 ( 29.1)
##              Some high school  382 ( 18.0)         396 ( 18.7)
##              High school      629 ( 29.6)         705 ( 33.2)
##              Some college      0 (  0.0)         0 (  0.0)
##              College          301 ( 14.2)         403 ( 19.0)
##  marital_status (%) Under 17      0 (  0.0)         0 (  0.0)
##              Married          1373 ( 64.7)        1547 ( 72.9)
##              Widowed          430 ( 20.3)         298 ( 14.0)
##              Divorced         128 (  6.0)         130 (  6.1)
##              Separated         83 (  3.9)          45 (  2.1)
##              Never Married     109 (  5.1)          97 (  4.6)
##              Blank, but applicable 0 (  0.0)          4 (  0.2)
##  urban (%)       Yes            1345 ( 63.4)      1278 ( 60.3)
##              No              778 ( 36.6)         843 ( 39.7)
##  alcohol (mean (SD)) 1.18 (2.75)  1.77 (3.54)
##  recex (%)       High            230 ( 10.8)         296 ( 14.0)
##              Medium           821 ( 38.7)         894 ( 42.1)
##              Low             1072 ( 50.5)         931 ( 43.9)
##  smoking (mean (SD)) 5.29 (10.78)  6.02 (11.71)
##  bmi (mean (SD))   26.72 (5.67)  25.97 (5.00)
##              Stratified by niacin_q

```

##		3	4	p	test
##	n	2122	2124		
##	sex (%)	1028 (100.0)	591 (100.0)	NaN	
##		0 (0.0)	0 (0.0)		
##	race (%)	1885 (88.8)	1931 (90.9)	<0.001	
##		200 (9.4)	156 (7.3)		
##		37 (1.7)	37 (1.7)		
##	age (mean (SD))	53.97 (13.32)	51.38 (13.09)	<0.001	
##	angina (%)	1665 (78.5)	1708 (80.4)	<0.001	
##		457 (21.5)	416 (19.6)		
##	arthritis (%)	1492 (70.3)	1608 (75.7)	<0.001	
##		630 (29.7)	516 (24.3)		
##	hypertension (%)	1466 (69.1)	1493 (70.3)	<0.001	
##		656 (30.9)	631 (29.7)		
##	aspirin (%)	1506 (71.0)	1588 (74.8)	<0.001	
##		616 (29.0)	536 (25.2)		
##	school (%)	517 (24.4)	418 (19.7)	NaN	
##		383 (18.0)	400 (18.8)		
##		779 (36.7)	791 (37.2)		
##		0 (0.0)	0 (0.0)		
##		443 (20.9)	515 (24.2)		
##	marital_status (%)	0 (0.0)	0 (0.0)	NaN	
##		1626 (76.6)	1719 (80.9)		
##		222 (10.5)	135 (6.4)		
##		120 (5.7)	100 (4.7)		
##		51 (2.4)	51 (2.4)		
##		98 (4.6)	117 (5.5)		
##		5 (0.2)	2 (0.1)		
##	urban (%)	1301 (61.3)	1303 (61.3)	0.211	
##		821 (38.7)	821 (38.7)		
##	alcohol (mean (SD))	2.61 (4.57)	3.73 (6.32)	<0.001	
##	recex (%)	326 (15.4)	408 (19.2)	<0.001	
##		947 (44.6)	1003 (47.2)		
##		849 (40.0)	713 (33.6)		
##	smoking (mean (SD))	7.70 (13.05)	9.56 (14.93)	<0.001	
##	bmi (mean (SD))	25.84 (4.87)	26.00 (4.51)	<0.001	